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Application of the CEEMDAN and stochastic resonance based unsaturated multi-stable symmetric stochastic feedback pooling network in bearing fault diagnosis

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ABSTRACT

Stochastic resonance (SR) is increasingly vital in bearing diagnostics for its capacity to enhance faint signals via noise exploitation, effectively converting noise into usable signal power. Nonetheless, current SR techniques display significant shortcomings in improving diagnostic capabilities. To overcome these obstacles, this study introduces an enhanced CEEMDAN-UMSSR-SFPN system, which incorporates the Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN) and Stochastic Feedback Pooling Network (SFPN) via optimized this setup using a Quantum Genetic Algorithm (QGA) to optimize the system parameters, leading to significant improvements. The key contributions of this study encompass the following: (1) A novel unsaturated multi-stationary symmetric potential function (UMSSR) has been introduced. This function merges sinusoidal and exponential components to broaden the scope of particle movement and effectively tackles the issue of output saturation; (2) The preprocessing method used by CEEMDAN for fault detection further enhances its applicability when processing complex non-stationary fault signals, ensuring stable performance under different operating conditions; and (3) The proposed method is incorporated into the Stochastic Feedback Pooling Network (SFPN) to develop a CEEMDAN-UMSSR-SFPN system, aiming to improve the accuracy of particle dynamics analysis and fault diagnosis. Experimental validation using the PADERBORN and Mechanical Failures Prevention Technology (MFPT) bearing fault datasets verify that the proposed system consistently delivers superior diagnostic performance across diverse operating conditions.

1. Introduction

Mechanical gear is now a staple across a myriad of sectors, from aerospace and railway travel to harnessing wind power [1]. How it runs safely and dependably can majorly sway societal calm and economic prowess. Hence, nipping early hiccups in the bud is crucial for keeping these systems up and running smoothly. However, complex and variable working conditions, along with harsh environmental factors, often result in monitoring signals with poor signal-to-noise ratios (SNR). Early micro-defect vibration signatures are often obscured by interference, complicating robust fault characteristic retrieval [2–4]. Conventional signal processing methods [5–7] often face inherent limitations in addressing these challenges. While aiming to suppress noise, such methods inevitably lead to the loss of useful signal components,especially when the signal is weak and noise interference is strong,

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reducing their effectiveness for early fault detection. Recent studies have further confirmed that under strong background noise and variable operating conditions, the weak fault-related impulses are easily submerged, and robust feature retrieval still remains challenging, motivating noise-assisted nonlinear enhancement strategies. SR as a representative nonlinear dynamical phenomenon, offers a novel and effective approach to signal enhancement. First introduced by Benzi et al. [8] in their study of the mechanisms behind Earth's glacial-interglacial cycles, SR typically arises in nonlinear systems subjected to appropriate external excitations [9]. Unlike traditional methods that aim solely to eliminate noise, SR emphasizes synergistic engagement between signal, noise, and the non-linear dynamics. In recent years, increasing attention has been paid to the constructive role of noise in intelligent fault diagnosis. Contrary to the conventional viewpoint that regards noise solely as a harmful disturbance, a growing body of research has demonstrated that appropriately introduced noise can act as a positive incentive to enhance weak feature perception in nonlinear systems. This concept, often referred to as positive-incentive noise or noise-assisted fault diagnosis, highlights that noise injection, when properly regulated, can significantly improve the sensitivity and robustness of fault detection systems, especially under low SNR conditions. Within nonlinear dynamical frameworks, noise can facilitate energy transfer, activate hidden system responses, and promote transitions that are otherwise inaccessible under purely deterministic excitation. As a representative nonlinear phenomenon, SR explicitly exploits the cooperative interaction among signal, noise, and system nonlinearity, enabling partial conversion of noise energy into useful signal energy. This mechanism has been widely validated in mechanical fault diagnosis, where weak periodic fault signatures are often submerged in strong background noise. Consequently, noise is no longer treated as an element to be completely suppressed, but rather as a beneficial resource that can be harnessed to amplify incipient fault features. By deliberately introducing noise and leveraging nonlinear characteristics of the system, SR is capable of transforming some noise into usable signal, thereby amplifying faint signals while suppressing background noise. This mechanism has not only been validated in natural phenomena such as biological systems [10] and atmospheric science [11], but also shows broad application prospects in engineering domains, including sensor design [12] and signal detection and enhancement [13]. In rotating machinery diagnostics, SR-based frameworks have been repeatedly reported to improve the detectability of incipient bearing faults and weak periodic impulses under low-SNR conditions.

Recent years have seen a notable focus on the application of SR in mechanical failure detection. Various enhanced SR models have been successively proposed and have demonstrated excellent performance in weak feature extraction. For instance, recent progress can be summarized from both data-driven monitoring and nonlinear enhancement perspectives. On the one hand, deep neural networks have been used to perform nonlinear compensation and longitudinal monitoring in optical links, demonstrating the effectiveness of learning-based strategies for complex nonlinear distortions [14]; meanwhile, unsupervised techniques have been exploited for intelligent anomaly detection systems, highlighting the practicality of label-efficient monitoring in real-world scenarios [15]. On the other hand, for mechanical fault diagnosis, adaptive vibration-assisted hybrid-potential cascaded multistable stochastic resonance has been reported to enhance weak bearing-fault features [16], and adaptive stochastic resonance in an improved Izhikevich neuron model driven by additive and multiplicative Gaussian noise has been applied to wind turbine fault diagnosis, indicating the potential of neuron-inspired SR in complex noise conditions [17]. These representative studies collectively motivate further exploration of robust SR-driven frameworks for weak bearing-fault feature enhancement. Methods such as scale-transformed SR [18,19], non-saturated potential models [20,21], and multi-scale noise modulation techniques [22,23] have all been shown to effectively amplify faint fault characteristic signals. As research has progressed, the scope of SR technology has broadened beyond the confines of mere bistable systems. It now encompasses more intricate and multistable arrangements. This expansion, achieved by incorporating multiple potential wells, has significantly enriched the dynamic behaviors of Brownian particles and facilitated transitions between these wells, in turn boosting the SNR of the entire system. Taking things a step further, Qiao et al. came up with a non-saturated potential well model designed to sharpen the detection of faint fault characteristics [24]. Jiang et al. devised a brand new two-dimensional quad-stable Gaussian potential SR model (TDQGSR) that handily curbs the output saturation problem often seen in coupled quad-stable models [25]. Meanwhile, He et al. pieced together a piecewise tri-stable SR (PTSR) method for teasing out fault signature signals, even with a lot of background noise [26]. Compared to conventional configurations, complex systems in the context of SR are increasingly designed through system-level coordination strategies, such as cascaded structures [27,28], coupled systems [29,30], array architectures [31], and the incorporation of feedback elements [32]. These approaches aim to form functionally coupled complex network systems and thereby achieve enhanced performance. This multi-system collaboration strategy is analogous to the signal processing approach employed in synthetic aperture radar, where multiple subsystems work in concert to detect weak signals more effectively. For instance, Zhang et al. introduced a cascaded system with a feedback structure, enabling the system to exhibit band-pass characteristics and thereby significantly enhancing signal amplification performance [33]. Building on previous research, Qiao et al. introduced improved approaches using fractional order array stochastic resonance systems [34] and coupled neuron stochastic resonance models [35]. These advanced methods have proven particularly effective for extracting fault characteristics, outperforming conventional techniques.

The idea of Stochastic Pooling Networks (SPN) originally emerged in 2007 work of Zozor, which framed them as a model for binary measurements derived from a shared data source across sensor arrays [36]. A defining characteristic of SPN lies in its pooling mechanism, which effectively suppresses random fluctuations and noise, thereby facilitating data compression and information extraction. SPN have since found extensive applications across a variety of domains, including analog-to-digital converter design [37], sensory systems including receptor cells and neural networks [38], along with communication networks [39].

In the context of SR applications, Shi et al. introduced a Weighted Segmented Bistable Stochastic Pooling Network (WSBSPN), incorporating a sparsity-based metric (GI) to evaluate its SR performance. This approach significantly enhanced the recognition of weak fault features [40]. Building upon this framework, Li and his team introduced the Modulated Periodic SR Pooling Network

(MPSPN), which leverages a normalized least-mean-square (NLMS) adaptive weighting strategy. By using the MPSR model as the sensory hub in the SPN architecture, the system significantly boosted the sensitivity of fault feature detection, as noted in Ref. [41].

It is worth noting that existing studies suggest a positive correlation between the count of stationary states within a SR system and its performance under certain conditions [42]. In this regard, He et al. introduced a novel method combining Adaptive Quasi-Periodic Potential Stochastic Resonance (QPPSR) with a Composite Blind Detection Index (CBDI), aimed at enhancing and detecting weak fault signals in mechanical systems such as bearings without relying on prior knowledge [43]. Periodic potential models [44,45], characterized by multistability and non-saturation properties, offer an infinite number of potential well pathways for Brownian particle transitions. This structure facilitates efficient energy transfer across potential wells, thereby further enhancing the capability for feature amplification of system.

The aforementioned research highlights the significant role of multi-system coordination mechanisms and diversified potential function designs in enhancing system performance. It is worth noting that both CEEMDAN-based preprocessing and SR-based enhancement share a common theoretical foundation, namely the deliberate utilization of noise to improve signal representation and feature extraction. In CEEMDAN, adaptive noise is intentionally injected into the decomposition process to suppress mode mixing and stabilize the extraction of intrinsic mode functions, thereby enhancing the reliability of non-stationary signal analysis. In contrast, SR-based methods leverage noise-induced transitions in nonlinear potential systems to amplify weak periodic components. Although these two strategies operate at different stages of the diagnostic pipeline, they are conceptually consistent in treating noise as a constructive factor rather than a disturbance. This noise-assisted paradigm provides a unified perspective for understanding the effectiveness of the proposed framework and further motivates the integration of CEEMDAN with SR based enhancement. Building upon this foundation, the present study proposes an improved diagnostic method, termed CEEMDAN-UMSSR-SFPN, designed to improve weak fault feature identification. Key contributions of the present study are outlined as follows:First, a new symmetric potential, possessing multiple stable states, results from merging a sinusoidal function with an exponential component. This design enhances particle dynamics and simultaneously addresses the typical output saturation challenges faced in conventional systems. Second, considering that bearing fault signals typically exhibit periodic impulsive characteristics. Thus, decomposition via CEEMDAN serves to refine initial signals. This step improves the accuracy and stability of subsequent feature extraction. To further optimize system performance, a quantum genetic algorithm (QGA) is utilized for adaptive tuning of critical parameters. On this basis, the proposed unsaturated multistable stochastic resonance (UMSSR) structure is integrated into a stochastic feedback pooling network (SFPN), resulting in the CEEMDAN-UMSSR-SFPN diagnostic system. This integration enhances both the transition efficiency of Brownian particles and the overall diagnostic capability of the system. Finally, experimental validations are conducted using multiple representative bearing fault datasets from the PADERBORN University and MFPT, in a variety of operating conditions. The findings validate that the proposed CEEMDAN-UMSSR-SFPN system exhibits superior performance compared to existing approaches in terms of fault identification robustness and SNR enhancement, demonstrating notable utility in applied engineering contexts.

2. Stochastic feedback pooling networks

2.1. Stochastic feedback pooling network model

Stochastic Pooling Networks (SPN) employ a network of nonlinear oscillators that operate in tandem, leveraging both random fluctuations and periodic input signals to filter out unwanted noise. By harnessing the synchronized behavior of these oscillators, the system dramatically reduces interference, effectively cleaning up the signal. This approach capitalizes on the inherent dynamics of the oscillator array to suppress noise [30], and the technique has been widely utilized across affiliated research domains. The conventional SPN system is described as follows:

$$\frac{d^2x_i}{dt^2} + r\frac{dx_i}{dt} = -\frac{dU(x_i)}{dx_i} + s(t) + \eta_i(t),\ \ i = 1, 2, \ldots, n \tag{1}$$

Where x_i denotes the displacement of ith oscillator, $s(t) = A\cos(2\pi f_0 t)$ is the period drive signal, $U(x)$ is the unsaturated multistable symmetry potential function, $\eta_i(t)$ denotes the Gaussian white noise, and the noises in each oscillator obey the Gaussian distribution and are independent of each other, which satisfy the following characteristics:

$$\begin{cases} \eta_i(t) = 0 \\ \eta_i(t)\eta_i(t+\tau) = 2D\delta(\tau) \end{cases} \tag{2}$$

The resultant system output is derived by calculating the mean of all oscillator output signals:

$$\langle x(t)\rangle = \frac{1}{n}\sum_{i=1}^{n} x_i(t) \tag{3}$$

Drawing from the principles of automatic control theory, feedback involves redirecting a portion or the entirety of a system outputs back into its inputs to optimize performance. Building on this concept, the Stochastic Feedback Pooling Network (SFPN) was developed to amplify sensitivity of the system to faint signals by incorporating individual feedback loops for every oscillator [46], as depicted in Fig. 1. The SFPN system operates as follows:

$$\frac{d^2x_i}{dt^2} + r\frac{dx_i}{dt} = -\frac{dU(x_i)}{dx_i} + s(t) + \beta x_i(t) + \eta_i(t),\ \ i = 1, 2, \ldots, n \tag{4}$$

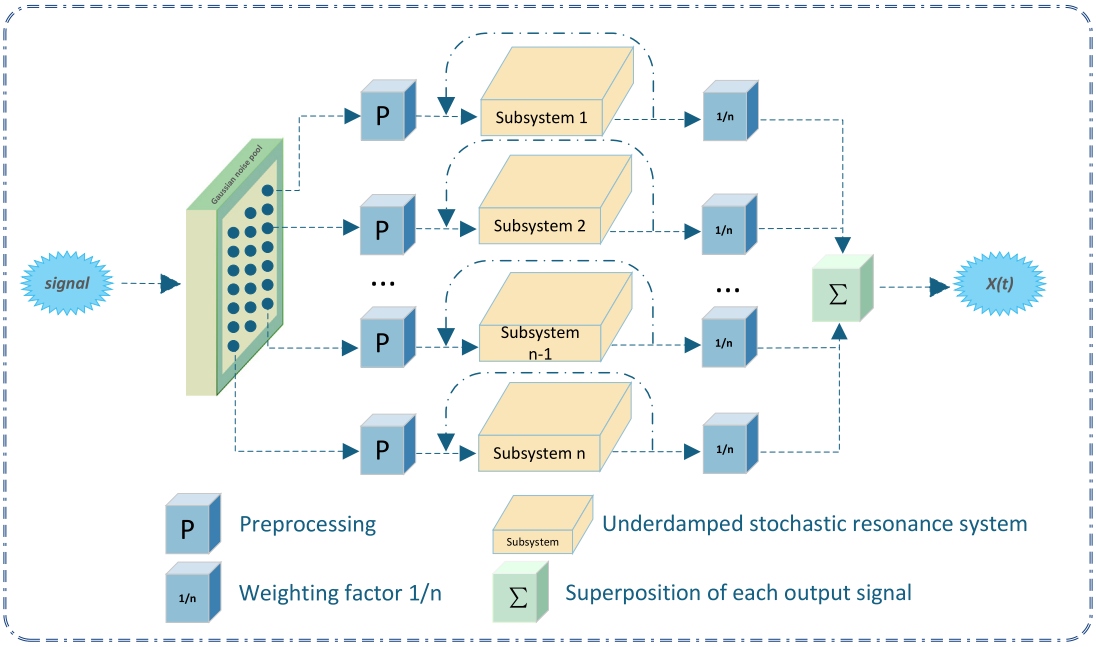


Fig. 1. Stochastic feedback pooling network model.

Here, r represents the damping factor, and β denotes the feedback coefficient. By introducing the feedback term βx_i, the system gain characteristics can be modulated, thereby enhancing its sensitivity to weak external inputs. The core operation mechanism of the stochastic feedback pooling network (SFPN) lies in the synergistic effect of feedback regulation and ensemble pooling. For each nonlinear oscillator in the network, the feedback term $\beta x_i(t)$ establishes a closed-loop regulation between the oscillator's output and input; when a weak target signal $s(t)$ is input, even if the initial response of the oscillator is extremely faint, the feedback term can dynamically amplify this weak response by adjusting the effective potential energy landscape of the system. Specifically, the feedback coefficient β modulates the system's equivalent stiffness: positive feedback ($\beta > 0$) enhances the oscillation amplitude of the oscillator in response to the target signal, enabling the weak signal to overcome the energy barrier of the potential well more easily; negative feedback ($\beta < 0$) suppresses irrelevant random fluctuations while maintaining the stability of the system's response to the target signal.

On this basis, the ensemble pooling mechanism further strengthens the extraction of weak signals. By averaging the outputs of multiple independent oscillators ($\langle x(t)\rangle = \frac{1}{n}\sum_{i=1}^{n} x_i(t)$), the random noise components in each oscillator (which are mutually independent and follow Gaussian distribution) are effectively offset due to their statistical randomness. In contrast, the target signal components, which are consistent across all oscillators, are superimposed and enhanced. This "noise cancellation, signal superposition" effect of the pooling process, combined with the dynamic amplification of weak responses by the feedback mechanism, significantly improves the system's SNR for weak inputs. Compared with the traditional SPN without feedback, SFPN achieves a dual enhancement of signal response sensitivity and noise suppression capability, making it more suitable for extracting faint fault features obscured by strong noise.

Moreover, a wealth of research has illustrated that boosting dynamic response of an oscillator can really improve how well a system performs. In contrast to the typical tristable and quad-stable potential well models, this paper puts forward an unsaturated multistable symmetric potential model (UMSSR) pieced together using sine and exponential functions to give dynamic behavior more variety of the oscillator, as demonstrated in Eq. (5).

$$U(x) = -a\sin^2(wx)\cdot e^{-b|x|} \tag{5}$$

The parameters a, b, and w in this study represent the deformation parameters of the potential function, where b and w are positive real numbers, respectively. Through modifying these parameters, the potential function can display various morphologies, thereby enriching the kinetic behavior of Brownian particles. As shown in Fig. 2(a), an increase in the parameter a results in a deeper potential well and a higher potential barrier, which significantly reduces the likelihood of Brownian particles transitioning between different potential wells. Conversely, Fig. 2(b) illustrates the opposite effect: as the parameter b increases, the potential wells gradually become shallower, and the potential barriers lower, making it easier for particles to overcome the barriers and transition across wells. Finally, Fig. 2(c) demonstrates the influence of the parameter w. As w increases, the width of the potential well narrows while its depth deepens, indicating that this parameter affects both the height of the potential barrier and the depth of the potential well. The overall trend in Fig. 2 reveals that the potential function $U(x)$ exhibits multiple steady states and is symmetric. Differing from the conventional model, confined to a finite set of stable equilibria [47], this system exhibits a theoretically infinite

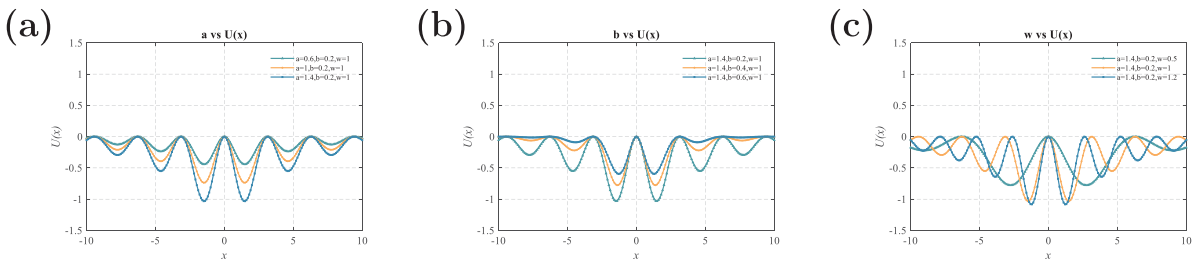


Fig. 2. Variation of the structure of the multistable potential function with system parameters.

number of such points. This characteristic overcomes the output saturation bottleneck encountered by the traditional model for the constraints of higher-order nonlinear terms, thereby significantly enhancing the system response capability. The height of the potential barrier plays a crucial role in determining the difficulty for particles to transition across the potential well. The higher the potential barrier, the greater the energy required for the particles to jump, which depends on the external cyclic driving force and the noise strength of the system. It is important to note that the jumping behavior of particles is closely related to the phenomenon of stochastic resonance, and changes in the jumping frequency directly influence the output performance of the system. Therefore, the careful selection and regulation of parameters are critical to achieving optimal response in the stochastic resonance mechanism.

For a clearer depiction of the proposed UMSSR system capacity to mitigate output saturation, we simulate its output response under cosine signals of varying amplitudes using the fourth-order Runge–Kutta algorithm. For comparison, the classical tristable stochastic resonance (CTSR) system [48], the empirically enhanced quad-stable stochastic resonance (CQSR) system [49], and the QPPSR system are selected to evaluate their performance with respect to output saturation. The specific potential functions of the selected comparison systems CTSR,CQSR and QPPSR are shown in Eq. (6),(7) and (8) respectively.

$$U_T(x) = \frac{\alpha}{2}x^2 - \frac{\mu}{4}x^4 + \frac{\gamma}{6}x^6 \tag{6}$$

$$U_Q(x) = -\frac{u^2v^2m^2}{2}x^2 + \frac{u^2v^2 + v^2m^2 + m^2u^2}{4}x^4 - \frac{u^2 + v^2 + m^2}{6}x^6 + \frac{x^8}{8} \tag{7}$$

$$U_p(x) = \frac{p(1-q^2)^2(1 + c\cos x)}{\left[1 + q^2 + 2q\cos\left(\frac{x}{d}\right)\right]^e} \tag{8}$$

The system parameters are set as follows: $a = 1.4$, $b = 0.2$, $w = 0.5$, $p = 0.6$, $q = 50$, $c = 2$, $d = 3$, $e = 2$, $\alpha = 1.45$, $\mu = 1.85$, $\gamma = 0.5$, $u = 2.6$, $v = 2.8$, $m = 3$; noise intensity $D = 0$; input signal $s(t) = A\cos(2\pi f_0 t)$, where $f_0 = 0.01$ Hz. The comparative simulation results are presented in Fig. 3, demonstrating the outstanding performance of UMSSR in mitigating saturation effects. It is observed that the output signal amplitudes of the CTSR and CQSR systems are constrained and exhibit only limited growth as the amplitude of the excitation signal increases. In contrast, as presented in Fig. 3(c), the output amplitude of the QPPSR system, a multistable potential system, exhibits a notable increase as the input signal amplitude rises. This behavior highlights the advantage of multistable systems over traditional systems with a finite number of steady states. More notably, the UMSSR system, depicted in Fig. 3(d), achieves even greater output enhancement compared to the QPPSR system, further underscoring its superior capability in mitigating output saturation.

2.2. Theoretical investigation

To assess the impact of adding feedback on the system, utilizing the ith oscillator as a model, we streamline the system dimensions to investigate the particles' dynamical responses. Let $\frac{dx_i}{dt} = y_i$, and then Eq. (4)can be transformed as:

$$\begin{cases} \frac{dx_i}{dt} = y_i \\ \frac{dy_i}{dt} = -ry_i - \frac{dU(x_i)}{dx_i} + \beta x_i + s(t) + \eta_i(t) \end{cases} \tag{9}$$

The Fokker–Planck equation governing this system is given by Eq. (10):

$$\begin{aligned} \frac{\partial\rho(x_i, y_i, t)}{\partial t} = &- \frac{\partial}{\partial x_i}\left[y\,\rho(x_i, y_i, t)\right] - \frac{\partial}{\partial y_i}\left\{\left[-ry - \frac{dU(x_i)}{dx_i}\right.\right. \\ &\left.\left. + \beta x_i + s(t)\right]\rho(x_i, y_i, t)\right\} + D\frac{\partial^2\rho(x_i, y_i, t)}{\partial y_i^2} \end{aligned} \tag{10}$$

This leads to the derivation of the steady-state probability density of the system as [50,51]:

$$\rho_{\mathrm{st}}(x_i, y_i, t) = N\exp\left[-\frac{\tilde{U}(x_i, y_i, t)}{D}\right] \tag{11}$$

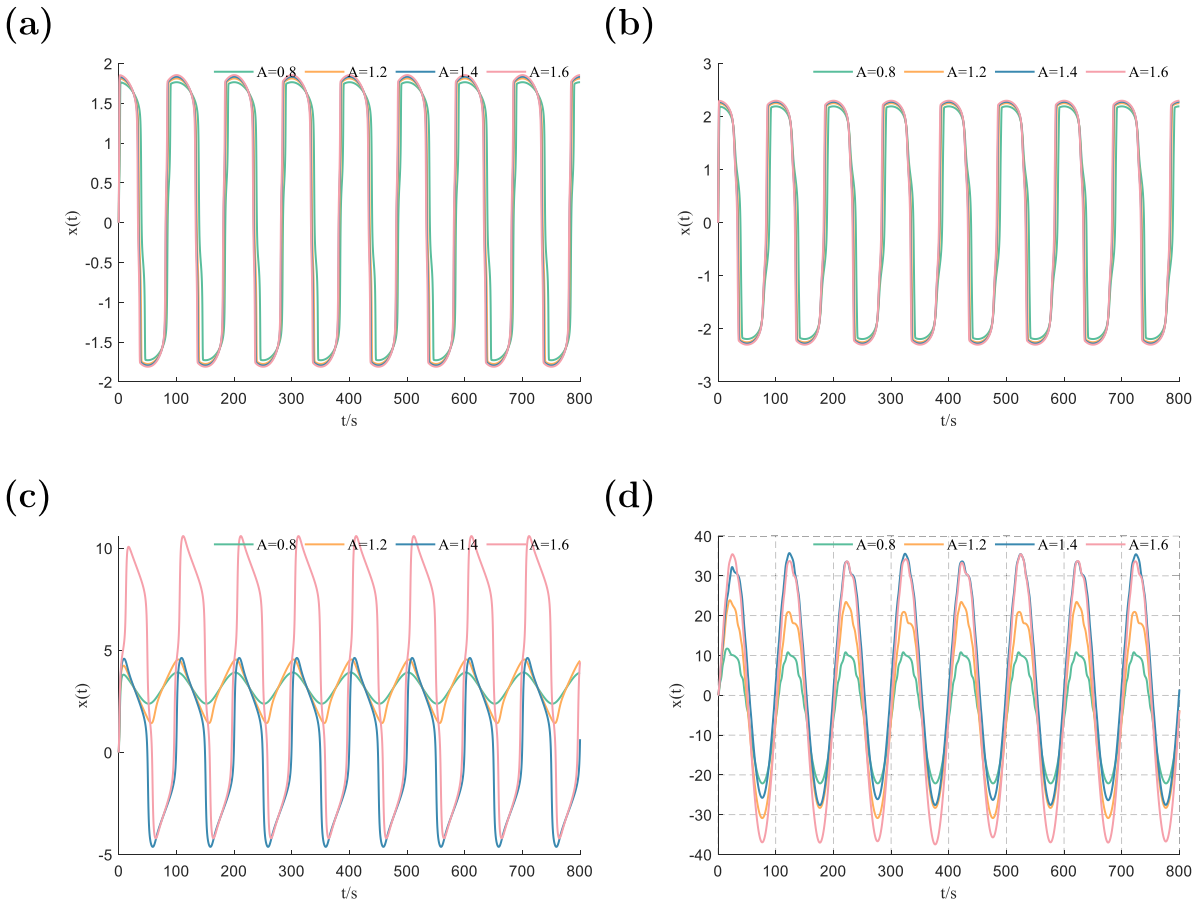


Fig. 3. Output of the (a) CTSR system, (b) CQSR system, (c) QPPSR system,(d)UMSSR system.

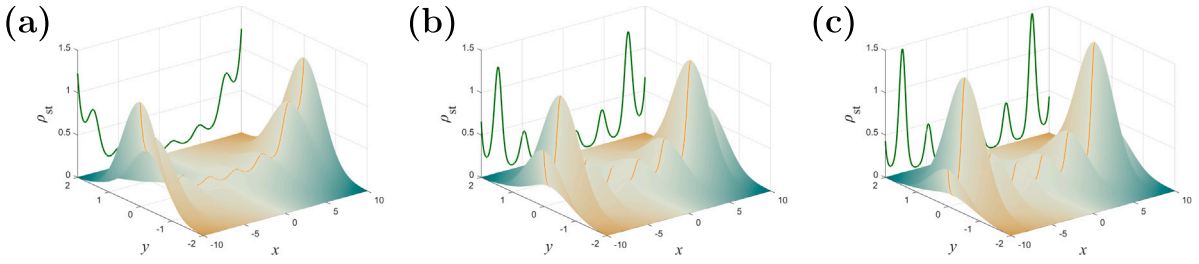


Fig. 4. Variation of steady state probability density with system parameters $a(b = 0.01, w = 1, r = 0.9, \beta = 0.02)$: (a) $a = 0.3$; (b) $a = 0.9$; (c) $a = 1.2$.

Where N is a normalization constant and $\tilde{U}(x_i, y_i, t)$ is a generalized potential function given by:

$$\tilde{U}(x_i, y_i, t) = r\left[\frac{1}{2}y_i^2 + U(x_i) - \frac{1}{2}\beta x_i^2 - x_i s(t)\right] \tag{12}$$

As depicted in Fig. 4, the probability of a particle remaining localized within each potential well gradually rises as the parameter a increases. This trend primarily arises from the deepening of the potential wells as a increases, which in turn raises the height of the potential barriers. The elevated barriers suppress the likelihood of Brownian particles transitioning between adjacent wells, thereby enhancing confinement within individual wells.

In contrast, Fig. 5 shows the opposite behavior: the potential wells become shallower as the parameter b increases, and the reduction in the height of the potential barrier promotes the transition of particles between different potential wells, thereby gradually reducing the likelihood of particles staying within the central potential well.

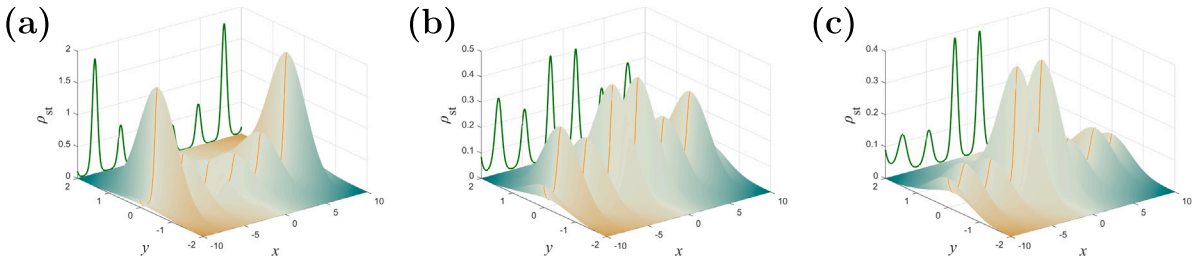


Fig. 5. Variation of steady state probability density with system parameters $b(a = 2, w = 1, r = 0.9, \beta = 0.02)$: (a) $b = 0.01$; (b) $b = 0.08$; (c) $b = 0.15$.

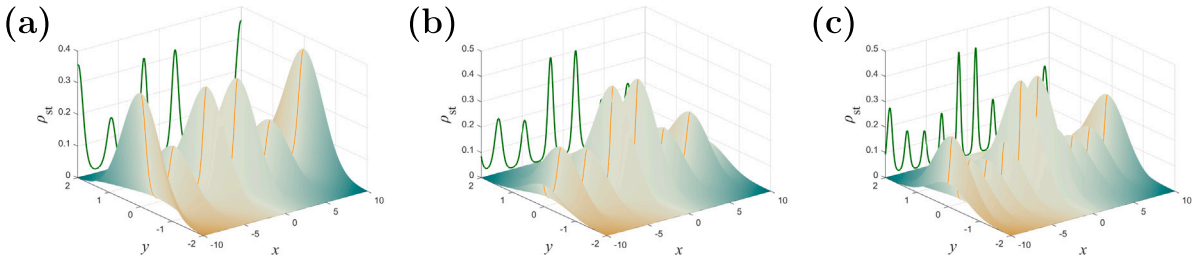


Fig. 6. Variation of steady state probability density with system parameter $w(a = 2, b = 0.1, r = 0.9, \beta = 0.02)$: (a) $w = 0.8$; (b) $w = 1$; (c) $w = 1.5$.

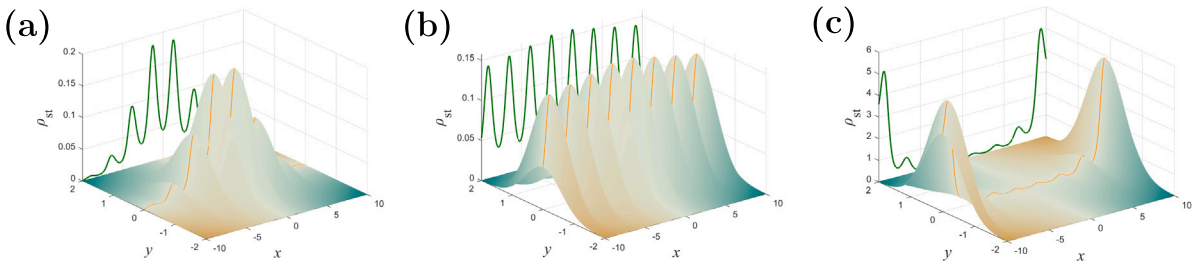


Fig. 7. Variation of steady state probability density with feedback coefficient $\beta(a = 2, b = 0.1, w = 1.5, r = 0.9)$: (a) $\beta = -0.03$; (b) $\beta = 0$; (c) $\beta = 0.03$.

Fig. 6 further illustrates the influence of the parameter w. As w increases, the potential wells not only deepen but also exhibit higher oscillation frequencies. These changes make particles more likely to be confined within the wells, particularly in the central region.

Fig. 7 presents the relationship between the steady-state probability density ρ_{st} and the feedback coefficient β. When $\beta < 0$ particles predominantly accumulate in the central potential well. Moreover, as β increases, the particle distribution gradually extends toward the lateral potential wells. This suggests the feedback mechanism enhances particle dynamic properties, offering greater possibilities for its distribution among the wells.

2.3. Numerical simulation validation

To assess the viability and efficacy of the developed UMSSR-SFPN system, computations employing the fourth-order Runge–Kutta technique were conducted [52]. The system is driven by a periodic signal of the form $s(t) = 0.01\cos(2\pi f_0 t)$, with each oscillator experiencing its own unique Gaussian white noise $\eta_i(t)$. The noise intensity is D = 0.2. The sampling process occurs at a rate of 5 Hz, and $N = 5000$ data points. In the field, the QGA has outshone other optimization algorithms in terms of efficiency, as noted in recent research [34]. Consequently, QGA utilized here to optimize system parameters, using SNR as the performance measure,as shown in Eq. (13). The algorithm uses a population size of 300 and a maximum iteration limit of 200. The parameter optimization boundaries are detailed as: damping coefficient $r \in [0, 1]$, feedback coefficient $\beta \in [-1, 1]$, system coefficients $a \in [-1, 1]$, $b \in [0.01, 2]$, and frequency parameter $\omega \in [0, 2]$. As the number of subsystems n increases, the average response of the system can be significantly enhanced, thereby improving overall performance; however, this also leads to increased system complexity. The search range for the number of system is set to $n \in [1, 50]$ considering the performance and the cost of computing.

$$SNR = 10\log_{10}\left(\frac{A_0}{\sum_{i=1}^{N/2} A_i - A_0}\right) \tag{13}$$

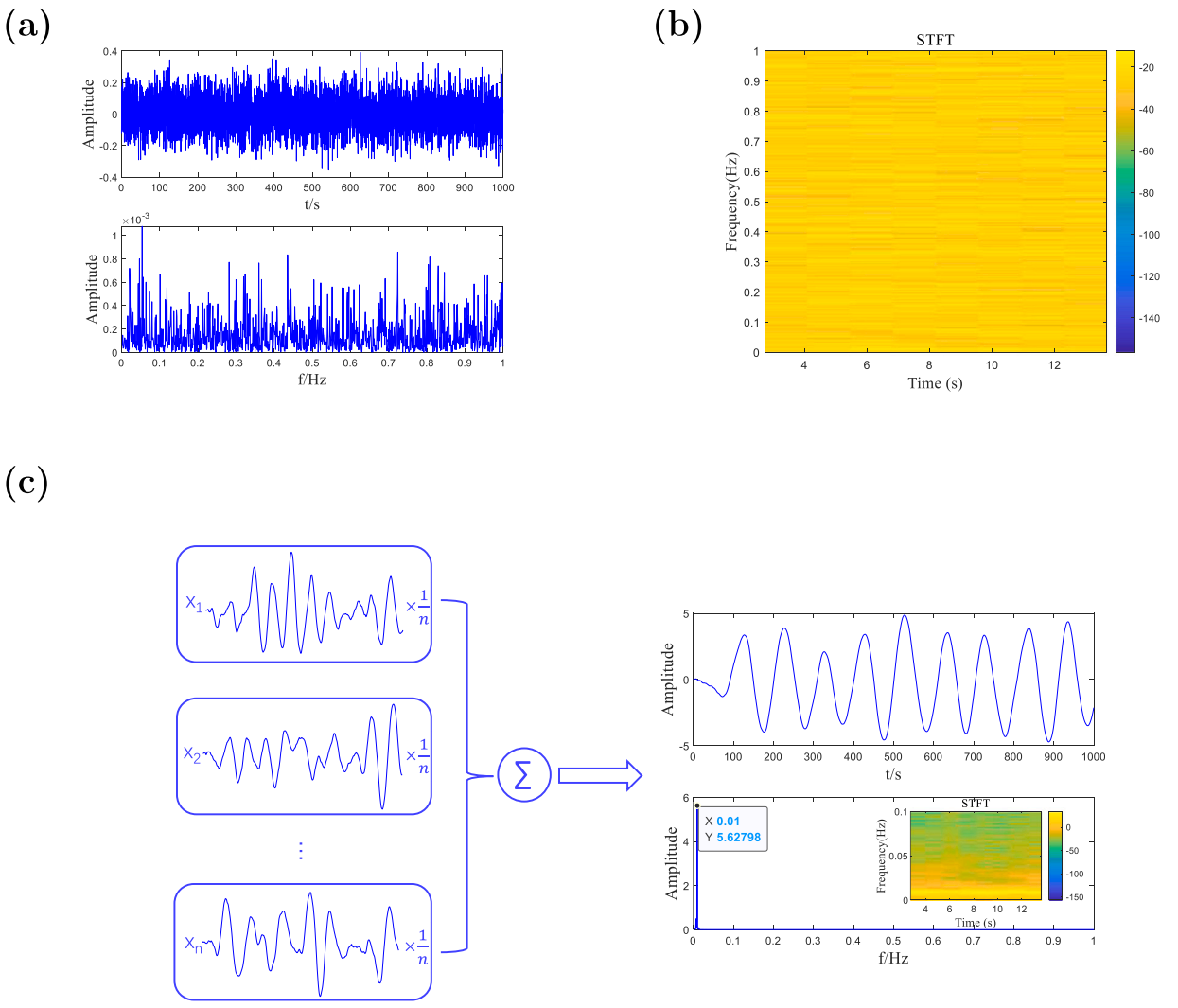


Fig. 8. (a) Time and frequency domains of signals containing noise; (b) Time-frequency diagram of a signal containing noise; (c) UMSSR-SFPN Processing and Output.

In Eq. (13), A_0 denotes the spectral amplitude of the system output at the target frequency f_0, while $\sum_{i=1}^{N/2} A_i - A_0$ represents the total spectral amplitude of the noise components excluding the target frequency. Due to the large number of subsystems, Fig. 8(a) presents the time and frequency domain representations of the noise-contaminated input signal on the first oscillator. It is found that the periodic components of the input signal are completely obscured by noise, rendering them nearly indistinguishable. Fig. 8(b) displays the corresponding time-frequency representation, where the target frequency component is similarly difficult to discern. In contrast, Fig. 8(c) illustrates the output response generated by the UMSSR-SFPN system architecture. By averaging the outputs of individual oscillators, random fluctuations are effectively suppressed, leading to enhanced system responsiveness and improved noise resilience. After applying QGA based parameter optimization, the following optimal parameter set was obtained: $a = 0.098527$, $b = 0.978788$, $r = 0.0395785$, $\beta = -0.0184736$, $\omega = 1.74535$, and system order $n = 37$.

To delve deeper into how the feedback coefficient and other parameters affect the performance of the UMSSR-SFPN system, we conducted multiple independent tests. These experiments employed identical noise conditions while maintaining all other variables consistent with the setup illustrated in Eq. (8). Fig. 9(a) depicts the correlation of average output SNR with noise intensity D as influenced by various feedback coefficient β. The results represent the mean SNR values obtained from 100 independent trials. All three curves exhibit nonlinearity, specifically rising in the initial stage and then declining, thereby confirming the presence of stochastic resonance within the system. Furthermore, Fig. 9(a) reveals how even minor adjustments to the feedback coefficient β can dramatically alter system performance, highlighting the extreme sensitivity of stochastic resonance systems to small parameter changes. Fig. 9(b) shows how the average output SNR varies with the number of subsystems n under different values of β. The results indicate that when n is relatively small, the system performance improves markedly with increasing n. However, as n continues to

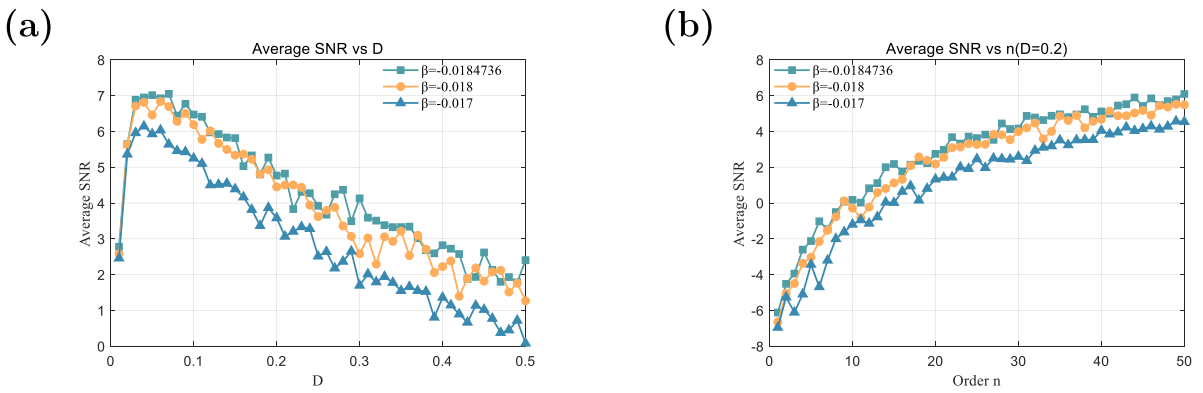


Fig. 9. (a) Variation in output SNR across various noise intensities for distinct feedback coefficients; (b) Variation in output SNR across various number of subsystems n for distinct feedback coefficients.

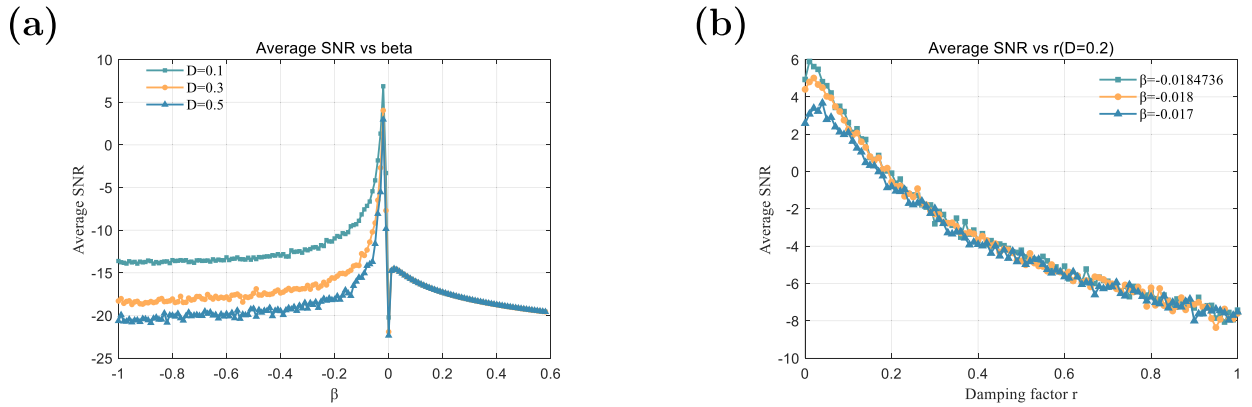


Fig. 10. (a) Variation in output SNR as a function of feedback coefficient β across various noise intensity; (b) Variation in output SNR as a function of damping factor r across various feedback coefficient β.

grow, the performance gain saturates and the curves level off. This suggests that in practical applications, the number of subsystems n should be appropriately selected to achieve a balance between complexity and performance.

Fig. 10(a) depicts the change in average output SNR with different feedback coefficients β under varying noise intensity. All three curves show a rise and fall trend, providing further evidence for the existence of generalized stochastic resonance. Due to numerical precision limitations, the SNR peak appears near $\beta = 0.02$. Theoretically, this result is consistent with the previously obtained optimal parameters in Fig. 8(a). Moreover, when $\beta = 0$, i.e., when no feedback is introduced into the system, the average output SNR is significantly lower than in the case with feedback, with the most pronounced improvement observed at $\beta = 0.02$, further highlighting the crucial role of the feedback mechanism in enhancing system performance. Fig. 10(b) presents the average output SNR versus damping coefficient r across varying feedback values. All curves reach their peak SNR values at relatively small r, which can be interpreted as follows: in the low-damping regime, the system approaches an underdamped state, thereby strengthening the SR effect and facilitating the transfer of noise energy into the output signal. Additionally, within this regime, the interaction among system parameters, noise, and driving signals achieves better coherence, enhancing the output SNR and forming a pronounced peak. As r increases further, this coherence deteriorates, leading to the suppression of resonance and a corresponding decline in SNR. Notably, when $\beta = -0.0184736$, the peak SNR occurs at $r = 0.04$, which closely matches the optimized result.

3. CEEMDAN-UMSSR-SFPN system

3.1. CEEMDAN

In practical applications, bearings typically operate under the influence of various interference factors, resulting in highly complex working conditions. Therefore, effective preprocessing of the raw fault signals is essential prior to further analysis. The Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN), an improved method based on EEMD, offers superior mode separation capabilities. It significantly enhances the accuracy and consistency of signal reconstruction while requiring fewer sifting iterations, thereby reducing computational cost effectively [53]. This study proposes the integration of the CEEMDAN

algorithm with the UMSSR-SFPN system. Initially, CEEMDAN is employed to extract frequency components from the raw signal, followed by the enhancement of key feature signals through the UMSSR-SFPN system, thereby enabling efficient fault source detection and identification. The decomposition process of CEEMDAN consists of the following main steps:

1. Add Gaussian white noise $w_i(t)$ with different random characteristics to the original signal $x(t)$ to construct multiple noise-contaminated sample signals:

$$x_i(t) = x(t) + \epsilon_0 w_i(t), \quad i = 1, 2, \ldots, I \tag{14}$$

In Eq. (14), $w_i(t)$ denotes a realization of Gaussian white noise with zero mean and unit variance. To ensure scale invariance across signals with different amplitudes, the noise amplitude is set proportional to the standard deviation of the input signal: $\epsilon_0 = \alpha\,\sigma_x$, where σ_x denotes the standard deviation of $x(t)$ and α is a dimensionless coefficient. In all bearing experiments, we fix $\alpha = 0.2$ and use an ensemble size $I = 100$ to ensure reproducibility.

2. Perform a single Empirical Mode Decomposition (EMD) on each noisy sample $x_i(t)$ to extract the first intrinsic mode function (IMF) component $\mathrm{IMF}_i^{(1)}$.
3. Compute the average of all first-order IMFs to obtain the first complete intrinsic mode function:

$$\widetilde{\mathrm{IMF}}^{(1)} = \frac{1}{I}\sum_{i=1}^{I} \mathrm{IMF}_i^{(1)} \tag{15}$$

4. Calculate the first-order residual signal:

$$r_1(t) = x(t) - \widetilde{\mathrm{IMF}}^{(1)} \tag{16}$$

5. Add scaled Gaussian white noise $\epsilon_k E_k(w_i)$ to the residual $r_1(t)$, and repeat steps 2–4 for each noise-perturbed sample until the residual signal becomes either monotonic or lacks sufficient extrema.
6. Finally, the original signal is reconstructed as the sum of all complete IMFs and the final residual:

$$x(t) = \sum_{k=1}^{K} \widetilde{\mathrm{IMF}}^{(k)} + R(t) \tag{17}$$

where $\widetilde{\mathrm{IMF}}^{(k)}$ denotes the kth complete IMF component, and $R(t)$ represents the final residual signal.

3.2. Signal reconstruction

To enhance the effectiveness of fault feature signals, this study employs a selection and reconstruction strategy for the IMF components obtained through CEEMDAN, based on the criterion of fault frequency band energy ratio. The specific procedure is outlined as follows:

1. For each signal, the power spectral density $P_k(f)$ of all IMF components is calculated, as defined in Eq. (18). Here, k denotes the kth IMF component, N represents the length of the component, and $\mathrm{IMF}_k[n]$ is the discrete-time sequence of the kth IMF. Particular attention is given to the spectral energy within the target frequency band (i.e., BPFO $\pm$ 5 Hz) to extract fault related features.

$$P_k(f) = \frac{1}{N}\left|\sum_{n=0}^{N-1} \mathrm{IMF}_k[n] \cdot e^{-j2\pi f n/N}\right|^2 \tag{18}$$

2. Select components based on the energy ratio within the target band. An IMF is retained if the energy within the specified band exceeds 0.5% of its total energy, indicating that it contains relevant fault features. If none of the IMFs meet this criterion, the IMF with the highest total energy is retained to avoid loss of critical information;
3. Linearly superimpose the selected IMFs to reconstruct the signal, which is then used for subsequent fault feature extraction and analysis.

This method fully leverages prior knowledge of energy distribution within the target frequency band, effectively enhancing the identifiability of fault signals. As a result, it provides more stable and discriminative input features for subsequent modeling and classification tasks. The CEEMDAN decomposition and signal reconstruction process is illustrated in Fig. 11.

4. Engineering applications

This section assesses the proposed method performance by bearing fault data. To evaluate system performance thoroughly, independent tests are performed on bearing data gathered across diverse operational scenarios. Comparative analyses are carried out against CTSR,CQSR and QPPSR systems, enabling a more objective and thorough evaluation of the proposed performance of the method.

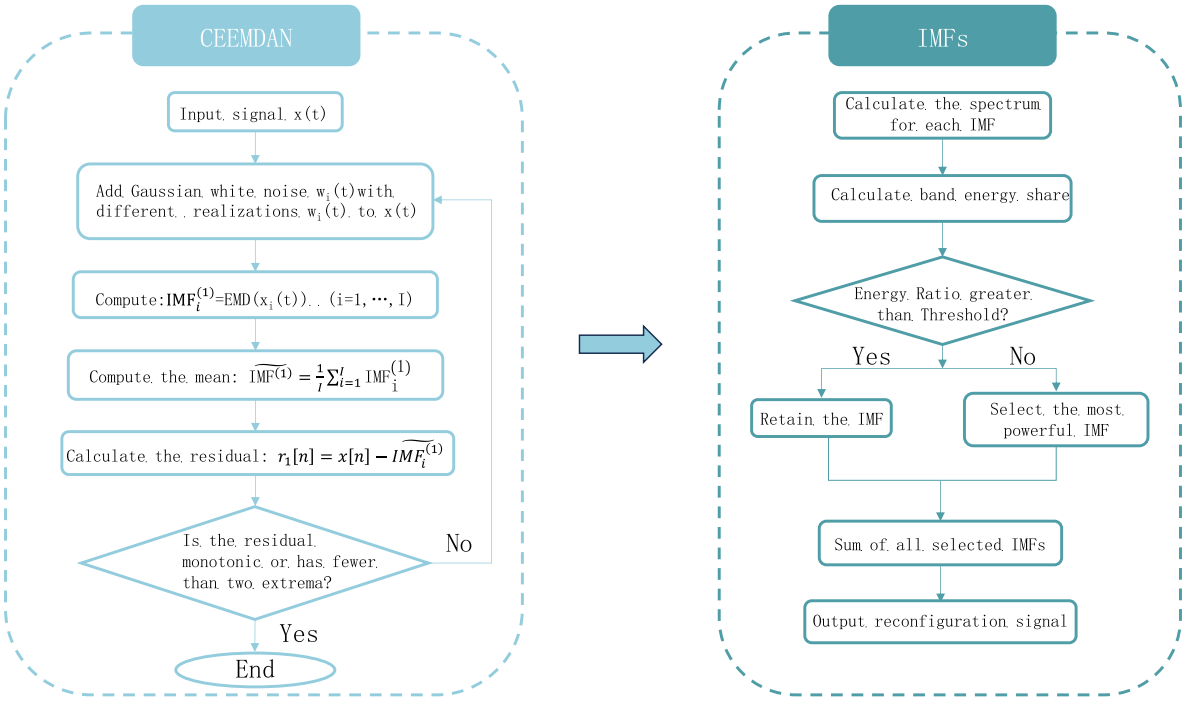


Fig. 11. CEEMDAN decomposition method and signal reconstruction flowchart.

Table 1
Structural parameters of the 6203 deep groove ball bearing.

Parameter	Parameter value	Unit
Pitch diameter D	29.05 or 28.55	mm
Number of RE n	8	pc
RE diameter d	6.75	mm
RE length	6.75	mm
Pressure angle	0	°

4.1. Case 1: Outer ring failure analysis based on PADERBORN dataset

In case 1, in the PADERBORN bearing dataset [54], an outer race fault signal is under examination. The test subject is a 6203 deep groove ball bearing, and the essential specifications are detailed in Table 1. Here, RE stands for the rolling elements. The experimental arrangement is depicted in Fig. 12. The chosen dataset is denoted as `N15_M07_F04_KA04_1`, originates from the actual outer race failure process recorded on the accelerated lifetime test platform, as illustrated in Fig. 13. Data was sampled at 64 kHz, with 1 s segment ($N = 64{,}000$ points) analyzed. Based on Eq. (19), the characteristic fault frequency was calculated to be $f_0 = 76.34$ Hz.

$$\mathrm{BPFO} = \frac{nf_r}{2}\left(1 - \frac{d}{D}\cos\alpha\right) \tag{19}$$

Due to the large number of subsystems, Fig. 14 shows the average values of the outer ring fault signals of n subsystems in both the time and frequency domains. As observed, the fault features are severely masked by strong noise, resulting in an average input SNR of only −38.99 dB. To optimize the parameters of each system, the QGA is employed, with the parameter search ranges defined as follows: CEEMDAN-UMSSR-SFPN and UMSSR-SFPN: $R \in [500, 10000]$, $a \in [-1, 1]$, $b \in [0.01, 2]$, $w \in [0, 2]$, $r \in [0, 1]$, $\beta \in [-1, 1]$, $n \in [1, 50]$; CTSR: $R \in [5000, 10000]$, $\alpha, \mu, \gamma \in [0.01, 3]$; CQSR: $R \in [100, 10000]$, $u, v, m \in [0.01, 3]$; QPPSR: $R \in [500, 10000]$, $p, c, d, e \in [0.01, 10]$, $q \in [0.01, 100]$. The above ranges are determined based on extensive experimental experience, where R denotes the scaling factor.

Fig. 15(a), (b), and (c) show the output signals processed by the CTSR, CQSR, and QPPSR systems, respectively. In contrast, Fig. 15(d) illustrates the output of the UMSSR-SFPN system without CEEMDAN. Among all the systems, UMSSR-SFPN demonstrates superior performance in terms of SNR and fault feature enhancement.

Fig. 16(a) shows the signal after CEEMDAN-based preprocessing, where noise is significantly suppressed and the SNR increases to −23.26 dB. Furthermore, Fig. 16(b) presents the output of the UMSSR-SFPN system using the CEEMDAN preprocessed signal as

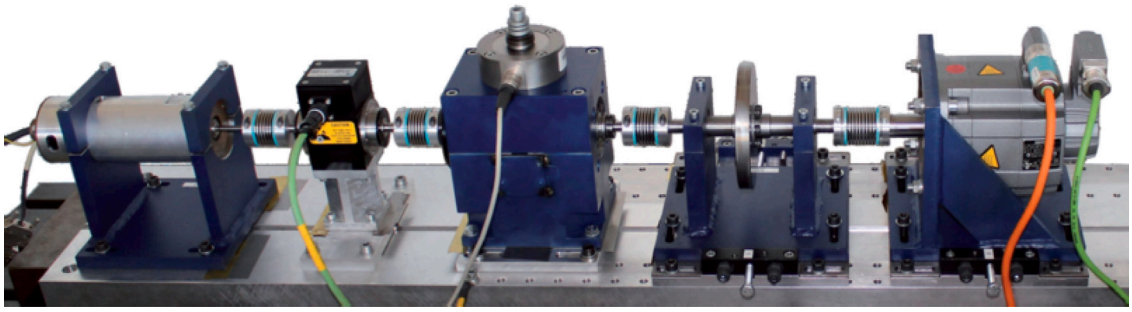


Fig. 12. Experimental setup for test bench.

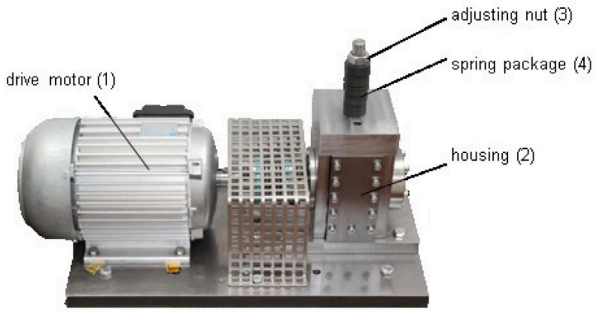


Fig. 13. Accelerated life test rig.

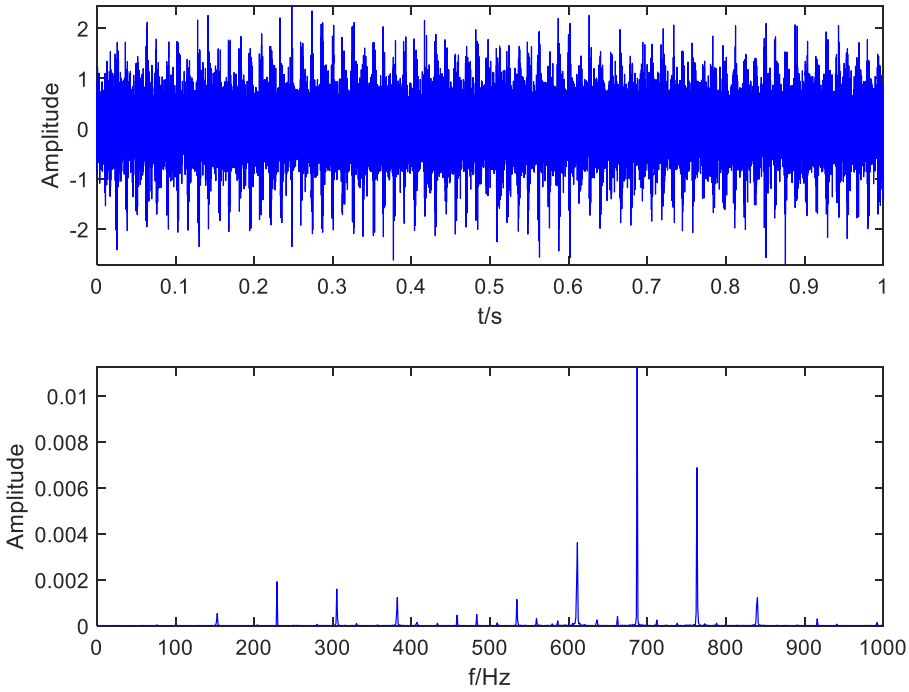


Fig. 14. Average the time and frequency domains of the input signal.

input, which exhibits better performance compared to the unprocessed case. Based on the above analysis and the data displayed in Table 2, the ranking of SNR improvement of different systems is as follows: CEEMDAN-UMSSR-SFPN > UMSSR-SFPN > QPPSR > CQSR > CTSR.

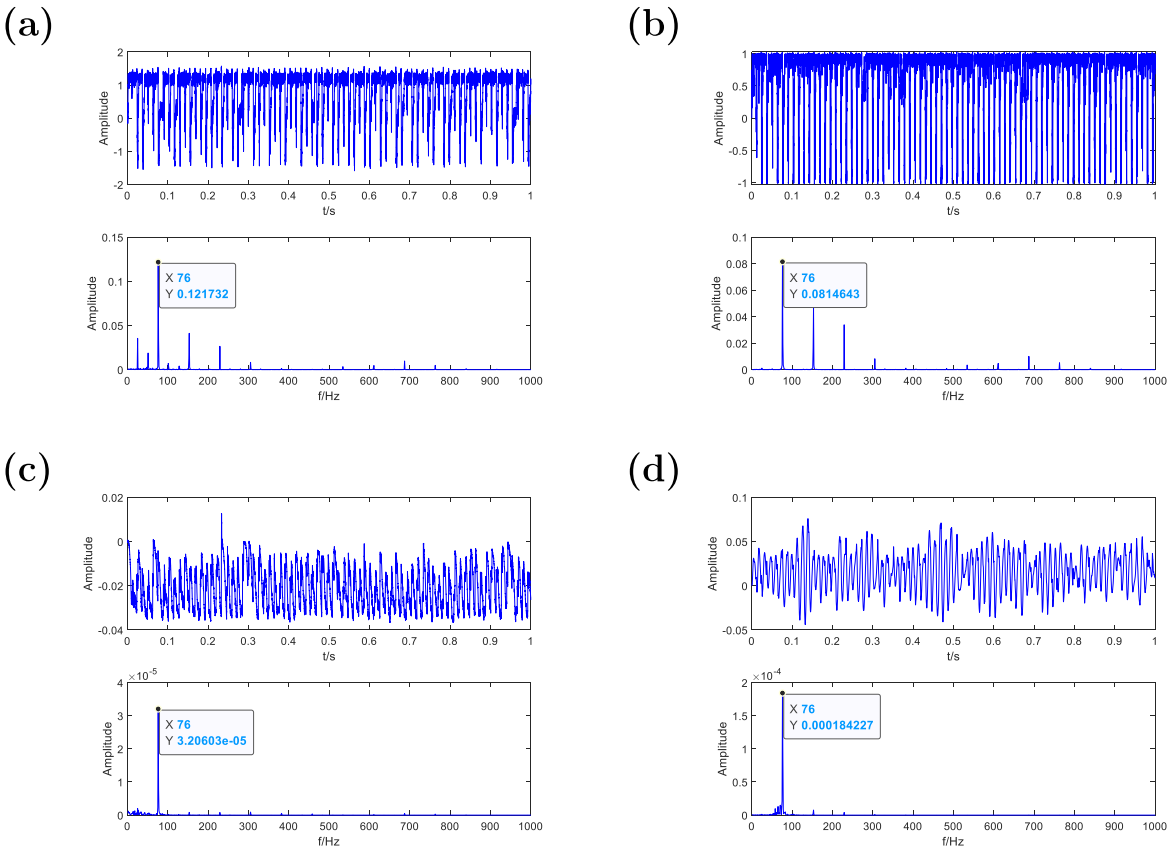


Fig. 15. Output results from (a) CTSR system, (b) CQSR system, (c) QPPSR system, (d) UMSSR-SFPN system.

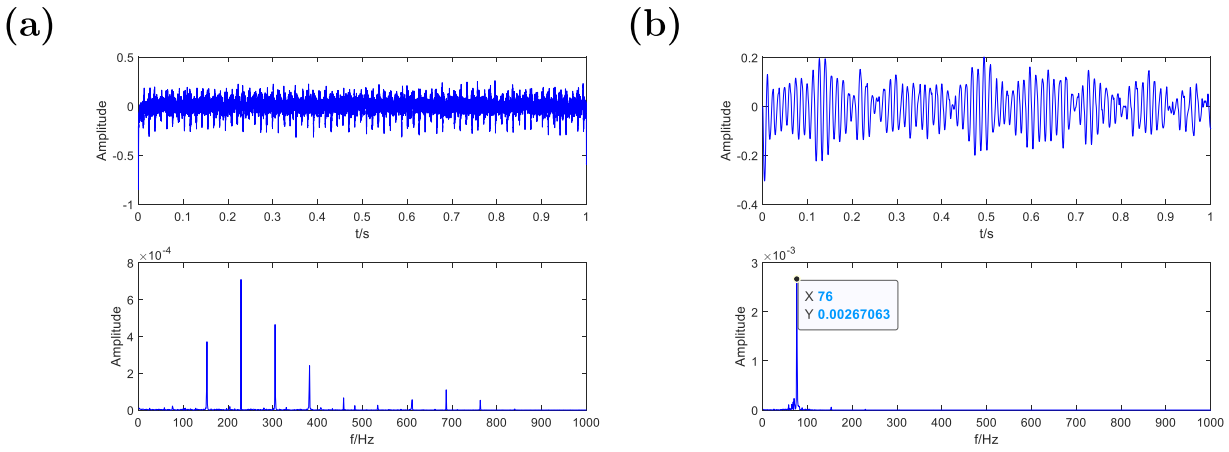


Fig. 16. (a) Reconstructed signal after CEEMDAN preprocessing; (b) CEEMDAN-UMSSR-SFPN system output signals.

4.2. Case 2: inner ring failure analysis based on PADERBORN dataset

In Case 2, further validation was conducted using real inner race fault data from the Paderborn dataset, specifically the file *N09_M07_F10_KI18_1*. The data were sampled at 64 kHz, and 1 s segment was extracted as the test signal. In accordance with Eq. (20), the fault characteristic frequency was calculated as $f_0 = 73.92$ Hz.

$$\mathrm{BPFI} = \frac{nf_r}{2}\left(1 + \frac{d}{D}\cos\alpha\right) \tag{20}$$

Table 2
Performance data and optimal parameter combinations for fault detection.

System type	Optimal parameter combination	SNR_{out} (dB)
CTSR	$R = 5810.1423$, $\alpha = 0.1772$, $\mu = 0.8852$, $\gamma = 0.5267$	−4.49
CQSR	$R = 6264.455$, $u = 0.4676$, $v = 0.1738$, $m = 0.8823$	−4.38
QPPSR	$R = 5055.9049$, $p = 0.0989$, $q = 15.2350$, $c = 0.4398$, $d = 0.3467$, $e = 0.7830$	−2.37
UMSSR-SFPN	$R = 1757.88765$, $a = -0.8290$, $b = 0.7499$, $r = 0.0786$, $\beta = -0.7431$, $w = 1.1862$, $n = 48$	−2.07
CEEMDAN-UMSSR-SFPN	$R = 2643.88385$, $a = 0.1437$, $b = 0.6525$, $r = 0.10785$, $\beta = -0.8196$, $w = 0.6683$, $n = 48$	−1.50

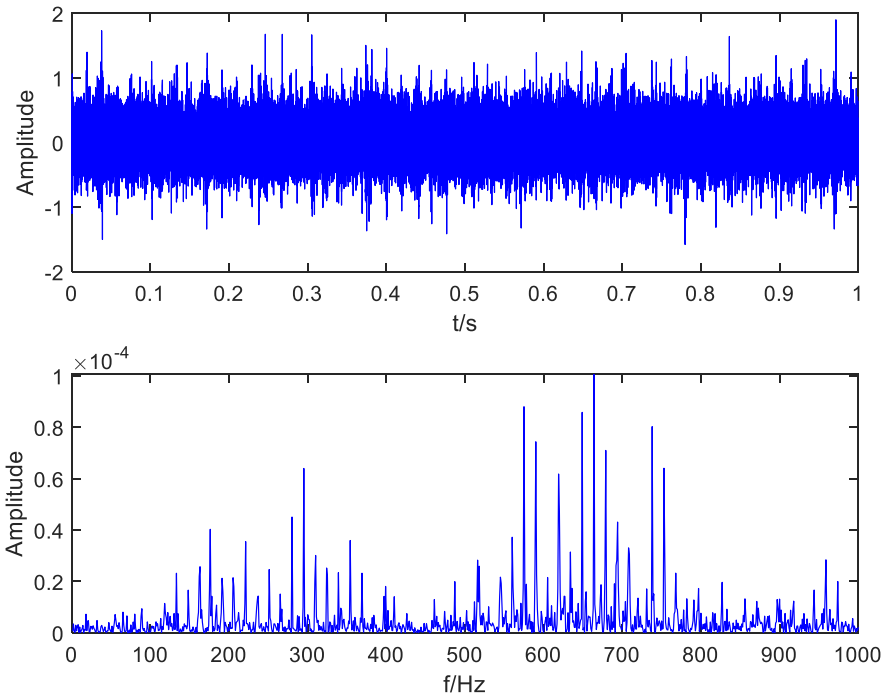


Fig. 17. Average the time and frequency domains of the input signal.

In rolling bearing fault diagnosis, the signal characteristics and diagnostic difficulty of outer race and inner race faults differ considerably. For the outer race, the defect position remains fixed, and periodic impacts are generated as the rolling elements pass over the fault. Consequently, outer race fault signals appear as regular impulses, with the characteristic frequency (BPFO) and its harmonics clearly visible in the spectrum and less affected by noise, making feature extraction relatively straightforward. By contrast, inner race defects rotate with the shaft, causing the impact excitations to vary with time. As a result, inner race fault signals often exhibit strong modulation effects, and their characteristic frequency (BPFI) is typically accompanied by rotational frequency components and harmonic sidebands. These features are more susceptible to noise and operating condition fluctuations, which increases the difficulty of feature extraction.

Fig. 17 displays the time and frequency domain representations of the unprocessed inner race fault signal, where the telltale signs of the defect are almost entirely drowned out by overwhelming background noise, with SNR as low as −60.74 dB. Despite the interference, the rotational frequency fr = 15 Hz and its harmonic elements remain discernible in the data.

Figs. 18(a), (b), and (c) present the output results of the CTSR, CQSR, and QPPSR systems, respectively. In these spectra, the rotational frequency and its harmonics are prominent, which prevents a significant enhancement of the fault features and makes their identification more challenging. In contrast, as shown in Fig. 18(d), the UMSSR-SFPN system was able to directly identify the fault features even without preprocessing.

Furthermore, Fig. 19(a) illustrates the time and frequency representation of the reconstructed signal after CEEMDAN preprocessing, where noise is effectively suppressed and the SNR is improved to −33.96 dB. When this preprocessed signal was fed into the

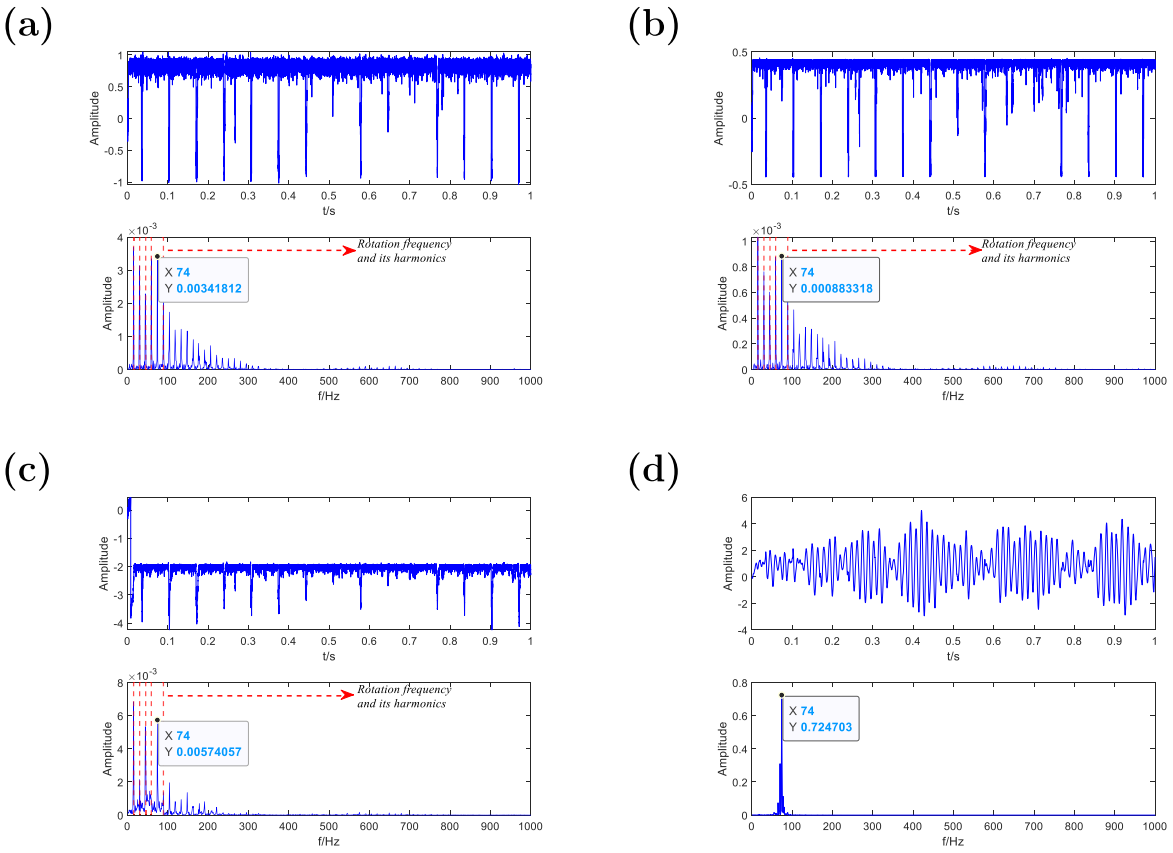


Fig. 18. Output results from (a) CTSR system, (b) CQSR system, (c) QPPSR system, (d) UMSSR-SFPN system.

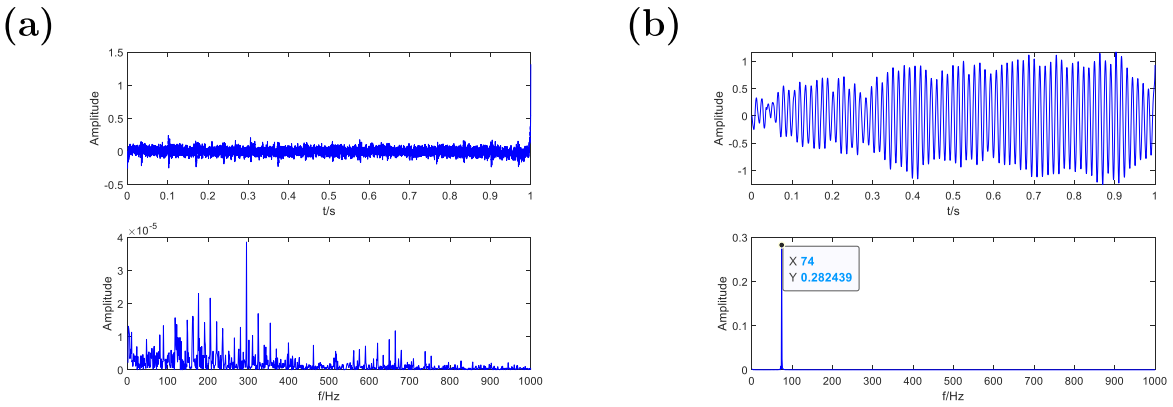


Fig. 19. (a) Reconstructed signal after CEEMDAN preprocessing; (b) CEEMDAN-UMSSR-SFPN system output signals.

UMSSR-SFPN system, the output results, shown in Fig. 19(b), demonstrated superior performance in both feature enhancement and SNR improvement compared with the unprocessed case. A detailed comparison of the different systems is summarized in Table 3.

4.3. Case 3: Outer ring failure analysis based on MFPT dataset

In Case 3, the bearing fault dataset provided by MFPT was employed for experimental analysis [55] . The MFPT test rig was equipped with a NICE type bearing, whose primary geometric parameters are as follows: roller diameter $d = 0.235$, pitch diameter $D = 1.245$, number of rollers $n = 8$, and contact angle $\alpha = 0°$. The *OuterRaceFault_1* dataset file was chosen for testing. Under an applied load of 270 lbs, an input shaft frequency of $f_r = 25$ Hz, and a sampling frequency of $f_s = 97,656$ sps, a vibration signal of total

Table 3
Performance data and optimal parameter combinations for fault detection.

System type	Optimal parameter combination	SNR_{out} (dB)
CTSR	$R = 9843.6736$, $\alpha = 0.29030992$, $\mu = 2.4392148$, $\gamma = 2.9777156$	−12.68
CQSR	$R = 5700.2329$, $u = 0.023977999$, $v = 0.18003999$, $m = 0.088011034$	−12.65
QPPSR	$R = 9419.6219$, $p = 2.1925707$, $q = 2.1332409$, $c = 0.59359911$, $d = 0.1964095$, $e = 8.8473784$	−12.52
UMSSR-SFPN	$R = 6527.6246$, $a = 0.41410772$, $b = 1.5394348$, $r = 0.01195718$, $\beta = -0.04937272$, $w = 0.48854874$, $n = 50$	−2.44
CEEMDAN-UMSSR-SFPN	$R = 3662.9521$, $a = 0.96823499$, $b = 1.93418$, $r = 0.011016856$, $\beta = -0.28126076$, $w = 1.7193277$, $n = 50$	6.94

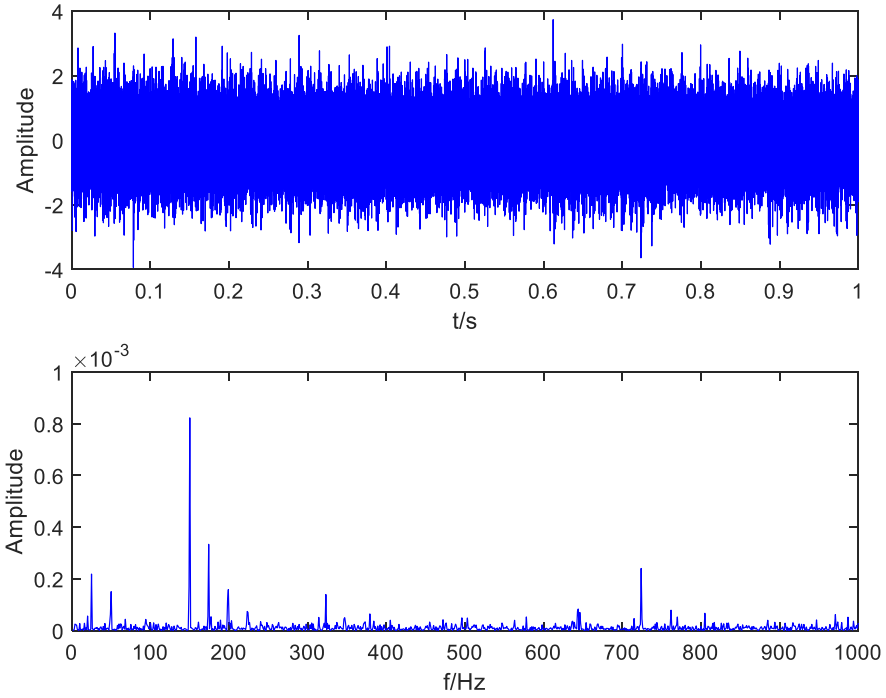


Fig. 20. Average the time and frequency domains of the input signal.

duration 6 s was acquired, from which 1 s segment was extracted for analysis. According to Eq. (19), the theoretical characteristic frequency of the outer race fault was calculated as $f_0 = 81.1244$ Hz.

Fig. 20 illustrates the time domain waveform and frequency spectrum of the original fault signal. Notably, the fault frequency component is heavily obscured by robust noise, complicating its straightforward recognition, with a SNR of only −43.99 dB.

Fig. 21 presents a comparative analysis of the time-frequency representations obtained after processing the signal with different SR systems. The results indicate that the multistable SR system significantly outperforms the classical tristable and quad-stable SR systems in enhancing the characteristic fault frequency, resulting in a more pronounced peak. Furthermore, compared with Fig. 21(c), the UMSSR-SFPN system not only yields more distinct fault features but also achieves a smaller detection error in the characteristic frequency.

To further enhance the feature extraction capability, the original signal was preprocessed using the CEEMDAN method. Fig. 22(a) shows the reconstructed signal after preprocessing, in which noise interference is significantly suppressed and the SNR is improved to −22.41 dB, resulting in a clearer representation of the fault features. Subsequently, the processed signal was fed into the CEEMDAN-UMSSR-SFPN system, and the time and frequency spectrum of it is illustrated in Fig. 22(b). Compared with the multistable stochastic resonance results without preprocessing (Fig. 21(d)), both the prominence and stability of the fault features are markedly enhanced, demonstrating the effectiveness of the preprocessing strategy. Optimization of all system parameters was performed using the QGA, and the optimal parameter configurations along with the corresponding performance evaluation metrics are summarized in Table 4.

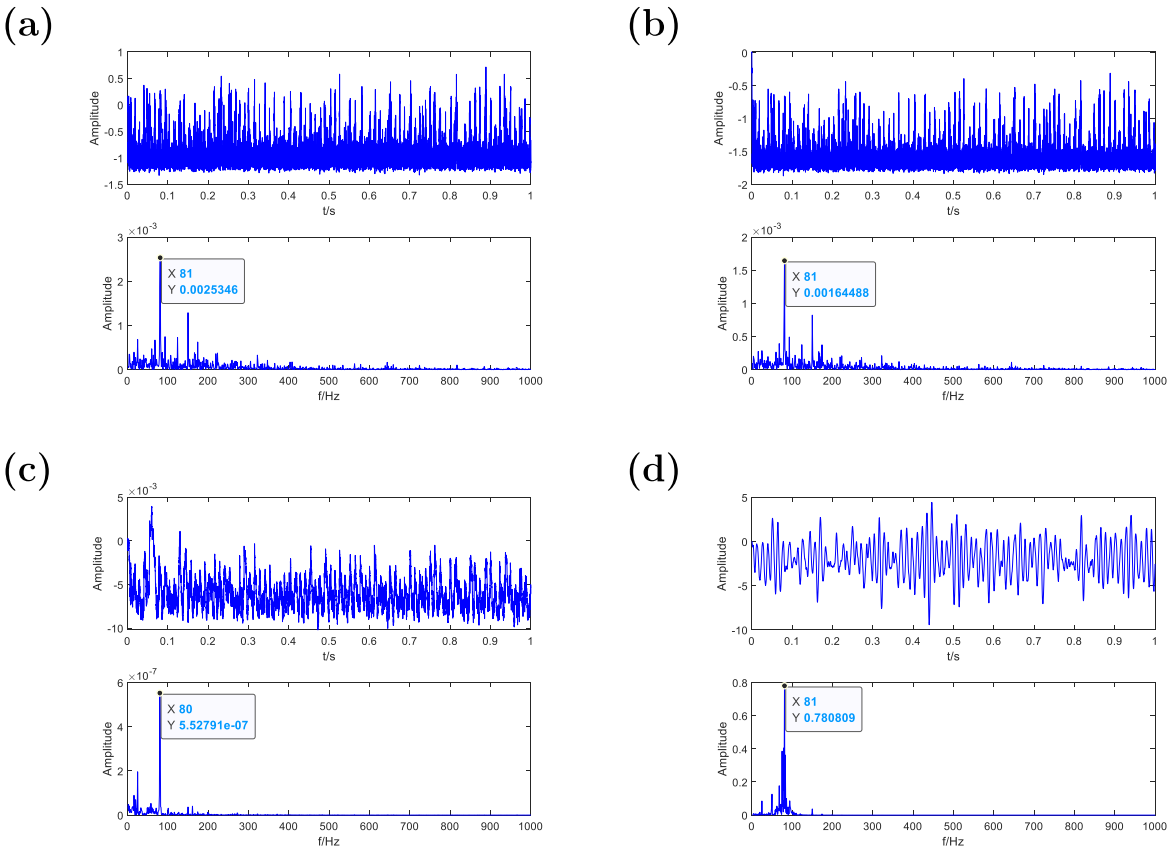


Fig. 21. Output results from (a) CTSR system, (b) CQSR system, (c) QPPSR system, (d) UMSSR-SFPN system.

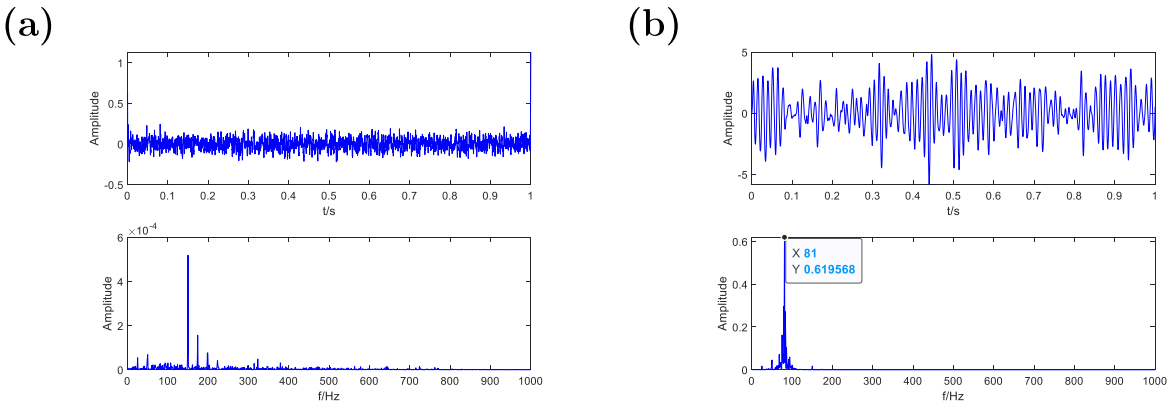


Fig. 22. (a) Reconstructed signal after CEEMDAN preprocessing; (b) CEEMDAN-UMSSR-SFPN system output signals.

4.4. Practical implementation in industrial settings

In an industrial deployment context, the proposed CEEMDAN-UMSSR-SFPN framework is designed to operate as a real-time streaming pipeline. Initially, vibration signals acquired from accelerometers or similar condition monitoring sensors are segmented via a sliding window mechanism. For each segment, CEEMDAN is employed to decompose the raw signal into Intrinsic Mode Functions (IMFs), after which a feature-enhanced component is reconstructed based on a band energy criterion centered around the characteristic fault frequencies (e.g., BPFO and BPFI). This reconstructed signal is subsequently processed by the UMSSR-SFPN module for weak-feature enhancement. Final diagnostic decisions can then be derived from the enhanced spectrum—specifically analyzing fault frequencies and harmonics.

Table 4
Performance data and optimal parameter combinations for fault detection.

System type	Optimal parameter combination	SNR_{out} (dB)
CTSR	$R = 7711.5609$, $\alpha = 0.63874475$, $\mu = 2.6273075$, $\gamma = 1.9886225$	−14.49
CQSR	$R = 6172.8214$, $u = 1.6219153$, $v = 2.6852384$, $m = 0.64530888$	−14.15
QPPSR	$R = 1498.7288$, $p = 0.11032793$, $q = 45.73692$, $c = 0.56539135$, $d = 0.23364563$, $e = 0.70267663$	−7.61
UMSSR-SFPN	$R = 7623.0589$, $a = 0.4551663$, $b = 1.1153542$, $r = 0.061451494$, $\beta = 0.055382781$, $w = 1.9823608$, $n = 46$	−6.42
CEEMDAN-UMSSR-SFPN	$R = 7145.748$, $a = -0.34569678$, $b = 1.0634479$, $r = 0.041827242$, $\beta = -0.070612975$, $w = 0.93674558$, $n = 46$	−5.34

The computational complexity of the system is primarily dominated by the CEEMDAN decomposition and the numerical integration of the n SR subsystems within the pooling network. Given that QGA-based parameter optimization is computationally intensive, it is recommended to perform this step offline using historical data. Consequently, online monitoring can employ fixed parameters or a finite set of condition-specific presets to ensure low latency. Furthermore, the mutual independence of the SR subsystems allows for parallel execution on multi-core CPUs or GPUs. It is also noted that the number of subsystems, n, should be optimized to balance diagnostic performance against the computational budget, as performance gains tend to saturate with large n.

Practical implementation faces challenges including variable operating conditions (such as fluctuating speeds and loads), intense background noise, impulsive interference, and variations in sensor placement. Additionally, the sensitivity of SR parameters to signal-to-noise characteristics suggests that periodic offline re-calibration or the development of condition-adaptive parameter tuning mechanisms will be essential for robust long-term operation.

5. Conclusion

Stochastic Resonance (SR), recognized for amplifying faint signals by leveraging noise, is extensively utilized in the realm of diagnosing rolling bearing failures. However, existing SR-based methods still face limitations in terms of the number of system steady states and the nonlinearity of the output response. To address these issues and improve overall system performance, this paper proposes a novel fault feature enhancement approach that integrates Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN), an unsaturated multistable symmetric potential function (UMSSR), and a stochastic feedback pooling network (SFPN), referred to as CEEMDAN-UMSSR-SFPN. The primary contributions of this study encompass the following:

1. To address the performance limitations of traditional stochastic resonance (SR) models caused by the finite number of stable states, a novel unsaturated multistable symmetric potential function (UMSSR) is proposed by combining a periodic sinusoidal function with an exponential term. This potential allows for flexible adjustment of system parameters to independently tune the barrier height, well depth, and steepness of the potential wells. As a result, it effectively expands the transition range of particles, alleviates the output saturation problem, and promotes inter-well transitions. These enhancements significantly improve the system capability for feature extraction.
2. Considering the pronounced periodic impulsive characteristics of rolling bearing fault signals, the Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN) is employed as a preprocessing step to extract and enhance key fault-related components while effectively suppressing noise interference.
3. Building upon this, the CEEMDAN decomposition results and the constructed UMSSR model are integrated into a Stochastic Feedback Pooling Network (SFPN), forming the CEEMDAN-UMSSR-SFPN framework to achieve deep enhancement of fault features.
4. Multiple sets of real-world fault condition data from the Paderborn and MFPT bearing datasets are used to conduct comparative experiments under various operating conditions. The experimental results demonstrate that the proposed method outperforms existing mainstream approaches in terms of feature extraction accuracy and fault identification performance, exhibiting strong robustness and practical applicability.

An important implication of this study is the re-examination of the role of noise in mechanical fault diagnosis. Rather than merely suppressing noise, the proposed framework demonstrates that noise, when properly regulated through adaptive decomposition, nonlinear dynamics, and feedback pooling mechanisms, can serve as a positive incentive to enhance weak fault signatures. This finding further supports the growing consensus that noise-assisted strategies are particularly suitable for intelligent fault diagnosis under complex operating conditions and severe background interference.

Although the proposed CEEMDAN-UMSSR-SFPN approach has demonstrated promising performance in fault-feature enhancement and identification, several limitations remain. As shown in Table 5, when applying the noisy signal from Case 2 to different

Table 5
Comparison of Methods in Terms of Workspace Memory and Time.

Methods	Workspace memory (MB)	Time (s)
CTSR	57.69	686.8661
CQSR	57.69	794.5638
QPPSR	57.97	2275.2698
CEEMAN-UMSSR-SPRN	813.36	18 989.4356

systems for processing, the method proposed herein significantly outperforms other individual systems in terms of both memory usage within the working area and runtime. First, the overall computational complexity is relatively high due to the inclusion of CEEMDAN decomposition and QGA-based parameter optimization, leaving room for improvement in deployment efficiency and response latency for online, real-time diagnostic scenarios. Second, the potential-function parameters in UMSSR exhibit a certain sensitivity to noise levels and signal characteristics. At present, parameter selection primarily relies on empirical ranges and validation-set tuning, and a unified adaptive parameter adjustment mechanism is still lacking. In addition, the performance stability of the proposed method under extremely noisy conditions and complex operating regimes requires further systematic verification.

Future work will focus on three directions: (1) optimizing the SFPN architecture by integrating lightweight network design and model compression strategies to reduce computational overhead and improve real-time diagnostic capability; (2) investigating parameter-adaptive mechanisms for varying operating conditions (e.g., online optimization strategies such as reinforcement learning) to enable dynamic UMSSR parameter tuning and enhance robustness; and (3) extending experimental studies and engineering validations to more challenging scenarios — including high-speed, heavy-load, and strong-noise conditions — to conduct a more comprehensive evaluation of the proposed method and verify its practicality and stability.

CRediT authorship contribution statement

Gang Zhang: Conceptualization. **Jiangmei Luo:** Formal analysis, Data curation. **Xiaoxiao Huang:** Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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